

0 5 . 1

Orchids form a large family of flowering plants. Scientists analysed genes coding for ribosomal RNA in orchids.

Explain how this analysis allowed the scientists to determine the phylogenetic relationships between species of orchids.

[2 marks]

0 5 . 2

Suggest **one** reason why analysing genes coding for ribosomal RNA allows phylogenetic relationships to be studied between all cellular organisms.

[1 mark]

0 5 . 3

Most orchid species are found in tropical rainforests. Two common features found in many of these species are:

- they grow up trees to reach the upper branches
- they only open their stomata at night.

Suggest how each of these **two** features benefits these orchids.

[2 marks]

They can reach the upper branches of trees _____

They only open their stomata at night _____



0 5 . 4

A student used an optical microscope to observe a piece of tissue from the lower surface of an orchid leaf.

The piece of leaf tissue observed was very thin.

Explain why this was important.

[2 marks]

0 5 . 5

The student produced a biological drawing of the leaf tissue they viewed through an optical microscope.

Give **three** ways the student could ensure they produce a correct biological drawing of the leaf tissue.

Assume the student uses a sharp pencil.

[3 marks]

1

2

3

10

Turn over ►

